Statistical Modelling & Analysis (STAT2004)
Applied Class (Week 4)

We will go through Question 1, 2 & 4 together in the Applied Class. Try the other questions yourself during the week and discuss your results with your tutor next week.

1. The number of trials required to obtain the first success is modelled by the geometric distribution, which has pmf
$$f(x;p) = p(1-p)^{x-1}, \qquad x = 1, 2, 3, \ldots \qquad p \in (0, 1].$$
It can be shown that if $X \sim \mathsf{Geometric}(p)$, then $\mathbb{E}_p(X) = p^{-1}$ and $\mathrm{Var}_p(X) = p^{-2}(1-p)$. Suppose $X_1, \ldots, X_n$ is a random sample from this distribution. Let p_0 denote the true value of the parameter.
   (a) Determine the expected information and observed information.
   (b) Suppose $p_0 \in (0, 1)$. State the asymptotic distribution of $\sqrt{n}(\hat{p} - p_0)$, where $\hat{p}$ is the MLE.
   (c) What is the distribution of $\hat{p}$ when $p_0 = 1$? Explain why this is not described by the usual asymptotic normality result.
   (d) Suppose you want to estimate the mean of the distribution, which is given by p^{-1}. Does the MLE of p^{-1} achieve the Cramer-Rao lower bound.

2. Suppose $X_1, \ldots, X_n$ is a random sample from a distribution with probability density function
$$f(x;\theta) = \tfrac{1}{2}(1 + \theta x), \qquad x \in (-1, 1), \qquad \theta \in [-1, 1].$$
Let θ_0 denote the true value of the parameter.
   (a) Determine the observed information.
   (b) Suppose $\theta_0 = 0$. Determine the expected information and state the asymptotic distribution of $\sqrt{n}(\hat{\theta} - \theta_0)$, where $\hat{\theta}$ is the MLE of θ.
   (c) Suppose $\theta_0 = -1$. Explain why the usual asymptotic normality result cannot hold for $\sqrt{n}(\hat{\theta} - \theta_0)$.

3. Recall that if $(N_0, N_1, \ldots, N_k)$ has a multinomial distribution with cell probabilities $(\pi_0, \pi_1, \ldots, \pi_k)$ and total count n, then
$$\mathbb{P}(N_0 = n_0, \ldots, N_k = n_k) = \frac{n!}{n_0! n_1! \cdots n_k!} \pi_0^{n_0} \pi_1^{n_1} \cdots \pi_k^{n_k},$$
where $n_0 + n_1 + \cdots + n_k = n$ and $\pi_0 + \pi_1 + \cdots + \pi_k = 1$. We will apply this distribution to the following problem. A certain type of plant cell is classified into four versions. A simple one-parameter model, motivated by genetic considerations, assigns the following probabilities to the four versions:
$$(2 + \theta)/4, \quad (1 - \theta)/4, \quad (1 - \theta)/4, \quad \theta/4,$$
where $\theta \in (0, 1)$ is unknown. The biological details are not important for this question; the aim is to estimate θ in this constrained multinomial model. Suppose a sample of n cells has the observed frequencies (n_0, n_1, n_2, n_3), where $n = n_0 + n_1 + n_2 + n_3$.
   (a) Show that, up to a multiplicative constant not depending on θ, the likelihood depends on n_1 and n_2 only through $n_1 + n_2$.
   (b) Does the MLE exist when $n_1 + n_2 = 0$? Justify your answer. Identify the necessary and sufficient conditions for the MLE to exist.
   (c) Determine an expression for the MLE of θ assuming it exists.
   (d) Determine the asymptotic variance of $\hat{\theta}$.

Adapted from Azzalini (1999) Statistical Inference - Question 3.19

4. In the Week 2 applied class, we used R to find the maximum likelihood estimator of the zero inflated Poisson distribution. Recall the zero inflated Poisson distribution has probability mass function

$$f(x;\pi,\lambda) = \begin{cases} \pi + (1-\pi)\exp(-\lambda) & x = 0 \\ (1-\pi)\exp(-\lambda)\frac{\lambda^x}{x!} & x = 1, 2, \ldots \end{cases}$$

where $0 \leq \pi \leq 1$ and $\lambda > 0$.

Adapt your code from last week to determine standard errors of the maximum likelihood estimators, and apply your code to the count data in the file `fish.csv`.

5. The inverse Gaussian distribution is often used to model positive, right-skewed waiting times, and it arises as the first passage time distribution of Brownian motion with drift. Let $X_1, \ldots, X_n$ be a random sample from an inverse Gaussian distribution with density

$$f(x;\mu,\lambda) = \left(\frac{\lambda}{2\pi x^3}\right)^{1/2} \exp\left\{-\frac{\lambda(x-\mu)^2}{2\mu^2 x}\right\}, \qquad x > 0,$$

where $\mu > 0$ and $\lambda > 0$. For this parametrisation,

$$\mathbb{E}(X) = \mu, \qquad \mathrm{Var}(X) = \frac{\mu^3}{\lambda}.$$

(a) Determine the score function for $\theta = (\mu, \lambda)$.

(b) Determine the maximum likelihood estimator of θ.

(c) Show that the expected information matrix is

$$I(\theta) = \begin{pmatrix} \frac{n\lambda}{\mu^3} & 0 \\ 0 & \frac{n}{2\lambda^2} \end{pmatrix}.$$

(d) What does it imply, asymptotically, that the expected information matrix is diagonal?